ADVANCED POWER BI

DASHBOARD OF USA SPENDING

ON KANSAS HEALTH AND

HUMAN SERVICES : 2020-2024

Project - 2: Haribabu Ambati (WSU id: w397r285)

BSAN-750: DATA VISUALIZATION || FALL 2025

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ABSTRACT

This project expands upon the earlier visualization work completed in Project 1 by enhancing a Power BI dashboard using advanced analytics and modern interactive visual components.

Building on the same USASpending.gov dataset of Health and Human Services (HHS) funding for Kansas (2020–2024), Project 2 integrates advanced KPI cards, a Funnel chart, and a Sankey diagram. The final output is a fully interactive dashboard with few advanced visualizations supporting analysts in understanding patterns of federal spending across Kansas

Part 1: Dataset Selection:

For this Project, Data was downloaded from the official USAspending.gov portal, the U.S. government's open data platform for federal contracts and financial assistance (Grants) to the state of Kansas from 2020 – 2024, individually each year as per respective web page design.

Regular Separate CSV files were extracted for each year (2020–2024) covering both Grants and Contracts, then combined in Power BI using Power Query, since both Grants and Contracts, individually by themselves, have same columns/labels/fields.

1.2 Variable Identification:

After merging all 10 files (Grants + Contracts for 2020 – 2024), the integrated dataset contains:

- Rows: 12,542 records
- Columns: ~7 core fields used for analysis out of many fields from both Grants & Contracts schemas.

Only these 7 fields are chosen which directly contribute to our planned DAX measures, visualizations, and relationships. Others are hidden in Power BI Model View.

Numerical & Categorical variables*:*

Numerical variables:

Field Name	Description / Business Meaning
federal_action_obligation	Dollar amount obligated by the federal government for a given award(Grant) or Contract (transaction-level spending value).

Note: This Numerical Variable is further used to calculate and create various DAX measures.

Categorical variables:

Field Name	Description / Business Meaning
Category	Derived helper column identifying whether a record belongs to **Grant** or **Contract** (based on source file name).
recipient_name	Recipient organization or entity receiving the federal award.
recipient_county_name	County in Kansas where the recipient is located (used for spending by county visual).
primary_place_of_performance_city_name	City where funded work or service is performed.
Source_file	Original CSV source (e.g., HHS_GRANTS_2023.csv) used for audit and category classification.

Used the Same variables from Project-1 as shown in above snip for this Project.

Part 2: Advanced KPI Cards:

Advanced KPI Upgradation (Using Chapter 18):

The KPIs were enhanced using the advanced layout and formatting techniques recommended in Chapter 18 of *Data Visualization with Microsoft Power BI*.

- **Multi-line KPI cards** now integrate key contextual values such as YoY Growth % and last year's spending, increasing insight density without adding clutter.
- **Conditional formatting** highlights positive trends in green and negative trends in red, making performance direction immediately visible.
- **Modernized card design** with rounded borders, subtle shadows, and strong text hierarchy improves readability and professionalism.
- **Dynamic slicer responsiveness** ensures KPI values update according to selected date ranges, months, quarters, counties, or award types, enabling precise drill-down analysis.

Overall, these enhancements significantly strengthen the analytical clarity and usability of the dashboard while preserving the KPIs' original purpose and value.

Figure 1: Advanced KPI Cards

Note: The empty space appears because only five years exist in the dataset, and the card visual uses a fixed "cards-per-row" layout and we used existing DAX formulas from DAX formulas of Project 1 to create these advanced KPIs.

Part 3: Choosing Visualizations:

Classic Visualization Explanations:

Column chart:

Column chart for "Spending by Year" is introduced in **Chapter 3**, which explains how column and bar charts are used for comparing values across categories. According to the **Visualization Compass in Chapter 2**, this chart aligns with the **Dynamics** principle because it shows trends over time with fewer than twelve periods. Since the dataset includes only five years (2020–2024), it fits this guideline perfectly. A column chart is appropriate because it clearly displays how annual HHS spending in Kansas has increased or decreased, helping users quickly interpret funding fluctuations and long-term trends.

Why the Column Chart Makes Sense:

- Fits Dynamics principle (<12 time periods).
- Highlights year-to-year changes.
- Supports understanding of long-term funding patterns.

Figure 2: Column Chart with Spending Overtime

Donut chart:

Donut chart for “Spending by Category (Grants vs. Contracts)” is covered in **Chapter 4**, which focuses on pie and donut charts for composition analysis. According to the **Structure**

region of the **Visualization Compass in Chapter 2**, donut charts work best with fewer than six categories. With only two categories—Grants and Contracts, the chart meets this rule. This

Visualization type is especially effective here because Grants represent roughly 97% of spending, while Contracts contribute about 3%, making the disproportion - easy to see immediately.

Why the Donut Chart Makes Sense:

- Matches Structure principle (<6 categories).
- Clearly communicates the distribution between grants and contracts.
- Effectively highlights the dominant share of grant funding.

Figure 3: Donut Chart with Spending distribution in ' % '

Matrix Chart (County-Level Spending & Awards):

The matrix used for “Total Spending” and “Average Award” is from **Chapter 12**, which explains that tables and matrices are ideal for **precise comparisons** and **row-level detail**. In line with the **Comparison** region of the Visualization Compass (Chapter 2), this visual allows users to easily evaluate spending across counties with exact values.

Why the Matrix Makes Sense:

• Supports accurate, row-by-row comparison of spending and award amounts.

• Conditional formatting (colors and data bars) enhances readability and highlights differences effectively.

• Provides detailed numeric insights that complement higher-level visuals like the donut and funnel charts.

County	Total Spending	Average Award
SHAWNEE	20,015,973,743.03	5,668,641.67
HODGEMAN	4,000,000.00	4,000,000.00
THOMAS	3,000,000.00	1,000,000.00
FORD	12,558,099.76	784,881.24
STANTON	1,500,000.00	750,000.00
LOGAN	15,306,802.75	588,723.18
SEDGWICK	332,555,753.39	526,195.81
Total	**22,558,898,184.72**	**1,798,668.33**

Figure 4: Matrix Chart with Colors and Conditional formatting

Advanced Visualization Explanations:

Funnel Chart (Spending Breakdown from Chapter 11):

The funnel chart follows the guidance from Chapter 11, which explains that funnels are designed for sequential stages where values decrease as you move downward. To align with this definition and address the earlier feedback, the funnel was restructured to show Total Spending, then its two major components—Grants Spending and Contracts Spending. This creates a valid step-down structure where each level represents a subset of the previous one.

Why the Funnel Chart Makes Sense:

• Matches the Structure rule for funnels by showing a hierarchical reduction (Total → Grants → Contracts).

• Clearly communicates how overall spending splits into its two major categories.

• Makes the proportional difference easy to interpret, especially since Grants dominate total spending.

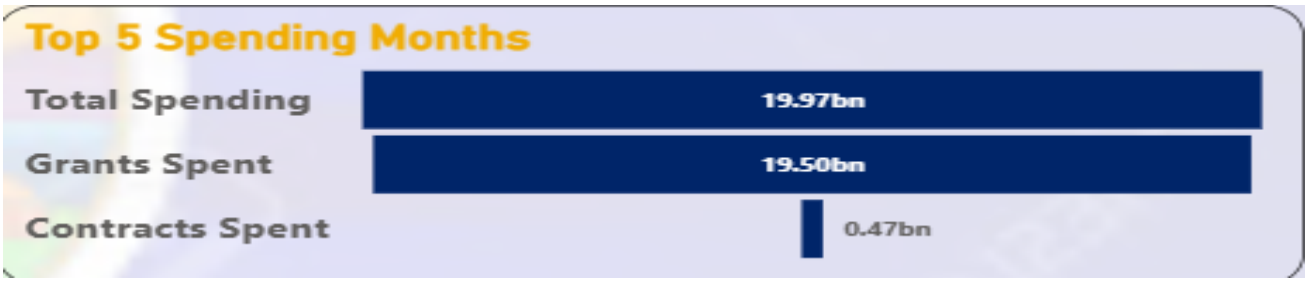


Figure 5: Funnel Chart Showing Conversion of Spending

<u>DAX Measure Snippets for Funnel Chart:</u>

```
Contracts Spending =
VAR ContractsTotal =
    CALCULATE (
        SUM ( USAspending_HHS_Kansas[federal_action_obligation] ),
        USAspending_HHS_Kansas[Source Type] = "Contract"
    )
RETURN
    COALESCE ( ContractsTotal, 0 )
```

Figure 6: Contract Spending DAX used in Funnel Chart

```
Grants Spending =
VAR GrantsTotal =
    CALCULATE (
        SUM ( USAspending_HHS_Kansas[federal_action_obligation] ),
        USAspending_HHS_Kansas[Source Type] = "Grant"
    )
RETURN
    COALESCE ( GrantsTotal, 0 )
```

Figure - 7: Grant Spending DAX used in Funnel Chart

Sankey Diagram (Funding Category Flow from Chapter 17):

The Sankey diagram follows **Chapter 17**, which recommends this visual for illustrating how values flow from one group to another. Here it shows how funding categories distribute across Grants and Contracts, with clear color coding (Contracts in red, Grants in green) to make patterns easy to interpret.

Note: The funding categories used in the Sankey were created using Power BI's **Grouping** feature and supporting **DAX logic** on the original Funding Agency field, allowing multiple agencies to be combined into meaningful, high-level groups. (Big DAX Formula and so couldn't screenshot it from formula box)

Why the Sankey Makes Sense:

• Shows how grouped categories split between Grants and Contracts.

• Color coding enhances readability and comparison.

• Aligns with Chapter 17's guidance for visualizing directional flow.

Figure 8: Colored Sankey Chart

Part 4: Dashboard Design:

The Power BI dashboard titled "USA Spending on Kansas Contracts & Grants (2020–2024)" is designed using a **3 × 2 grid layout**, selected to support a clean top-down analytical flow (Overview → Comparison → Details) while keeping all visuals evenly spaced and easy to navigate. Visual alignment was maintained using Snap-to-Grid and Snap-to-Shape, ensuring consistent positioning across all rows and columns.

Design Choices:

- *Grid Used:* A structured 3 × 2 layout was chosen to clearly separate KPIs, comparison visuals, and detailed insights into three readable levels.
- *Alignment:* Snap-to-Grid and Snap-to-Shape settings ensured consistent spacing and perfect alignment of visuals across the dashboard.
- *Background & Shapes:* A custom background was created by combining the USA and Kansas flags using AI. This background was softened for readability, and rounded rectangle shapes were used to group visuals. The dashboard's color palette was derived from the Kansas flag—blue (002569) and yellow (F1AD02)—and applied to titles, labels, and key chart elements for consistent branding.

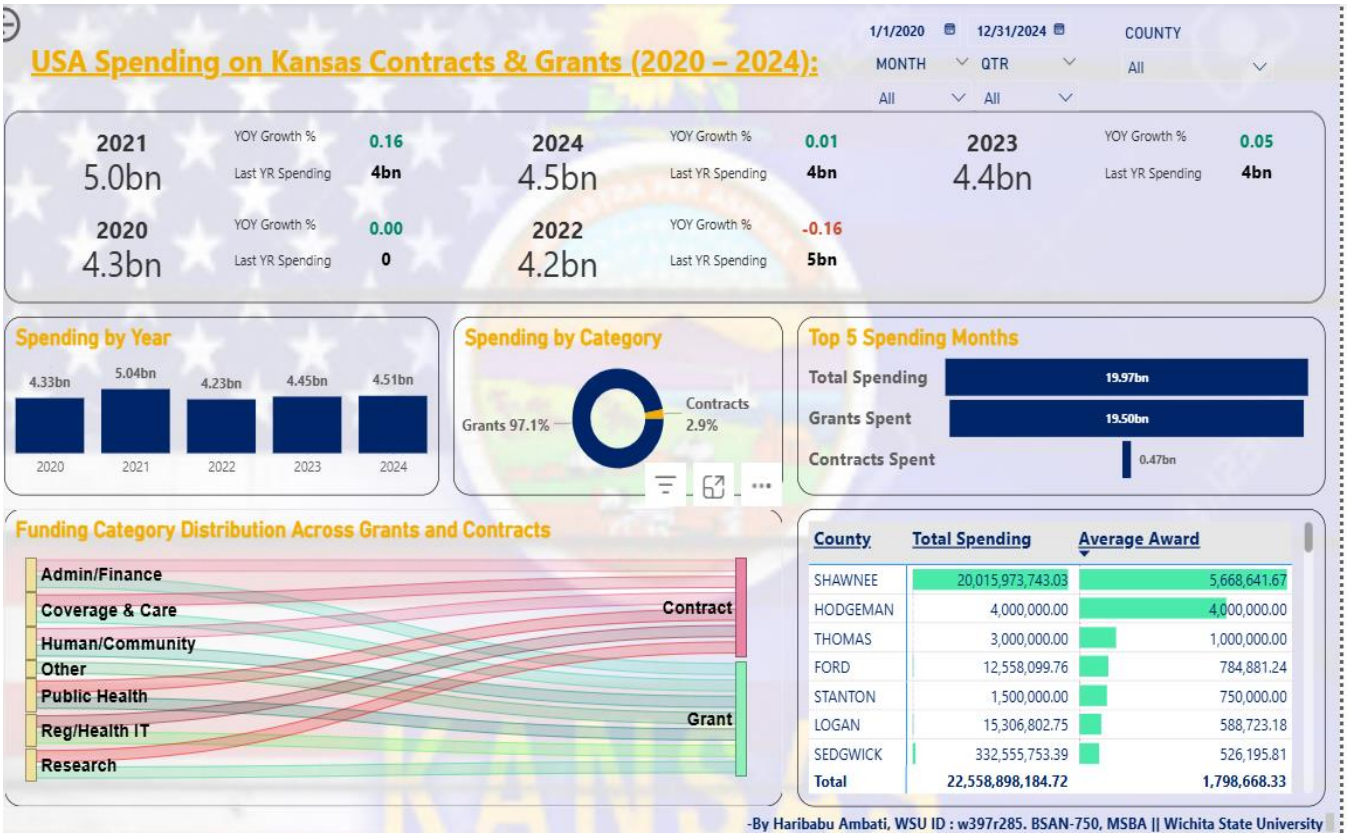


Figure 9: Final Dashboard

Level 0 - Header Area:

The top of the dashboard includes the main title **"USA Spending on Kansas Contracts & Grants (2020–2024)"** and global slicers for date, month, quarter, and county. This level establishes the scope and allows users to dynamically adjust the view before exploring the visuals.

Figure 10: Level 0 - Header Area

Level 1 – Advanced KPIs:

This level presents five years of spending advanced KPIs, YOY Growth %, and last year's benchmarks. Following Chapter 10's *Overview First* principle, this section delivers the most essential information immediately and helps users understand overall performance briefly.

Figure 11: Level 1 - Advanced KPIs

Level 2 - Summary Visualizations:

The second level contains three comparison-focused visuals arranged side by side:

- **Spending by Year (Column Chart)** – shows how total federal spending changes from 2020–2024.
- **Spending by Category (Donut Chart)** – compares the share of Grants versus Contracts.
- **Spending Breakdown (Funnel Chart)** – starts with Total Spending and narrows to Grants and Contracts, emphasizing how overall funds are allocated across the two types.

Together, these panels follow the **"Compare Next"** guideline by summarizing how spending differs across years and funding types before moving to more detailed views.

Figure 12: Level 2 - Summary Visualizations

Level 3 - Detailed Information:

This final level provides deeper, drill-down analysis through two detailed visuals. The Sankey diagram reveals how major funding categories flow into Grants and Contracts, while the county-level matrix provides precise spending and award comparisons. These visuals align with the *"Details Last"* guideline by giving users the most granular insights at the end of the dashboard.

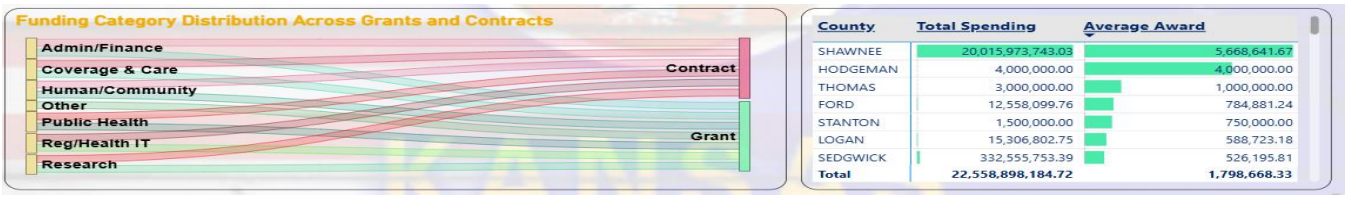


Figure 13: Level 3 - Detailed Information

Sankey Diagram:

- Shows how grouped funding categories distribute between Grants and Contracts.
- Color-coded flows help users quickly identify category-wise allocation patterns.

County-Level Matrix:

- Provides exact spending and award values for each county with conditional formatting.
- Enables clear comparison of county-level performance through data bars and color cues.

Part 5: Dashboard Pitch:

This dashboard is designed for **public-sector executives, grant managers, policy analysts, and nonprofit leaders** who rely on federal funding insights to make budget, planning, and compliance decisions. It organizes five years of Kansas HHS spending into a clear, interactive view, enabling stakeholders to monitor funding patterns, assess distribution fairness, and understand how grants and contracts flow across agencies. Universities, healthcare institutions, and county-level officials can also use this dashboard to evaluate how their region compares to others in attracting federal dollars.

The dashboard addresses a core business problem: **lack of unified visibility into federal HHS spending**. Instead of manually analyzing thousands of transaction records, users get a consolidated, filter-driven interface that reveals trends, category distributions, top performance periods, and funding pathways. Advanced KPI cards summarize overall performance, funnel charts highlight monthly spending peaks, and Sankey flows show how funding agencies distribute money across grants and contracts. This turns fragmented data into actionable intelligence.

Users interact with the dashboard by applying slicers—such as year, quarter, month, county, and funding category—to drill into specific time periods or regions. Each visual updates instantly, enabling comparisons, spotting anomalies, and tracing fund movement. The dashboard is valuable because it supports **evidence-based budgeting, program evaluation, and strategic planning**, helping organizations justify funding requests, forecast revenue, and identify areas

needing attention. In short, this dashboard transforms raw federal data into a decision-making asset worth investing in.

New Insights:

Insight 1 - YOY Growth Shows Recovery After a 2022 Decline:

Insight Statement: The advanced KPI cards reveal that 2022 experienced a **–16% decline**, but spending rebounded in 2023 and 2024 with **positive YOY growth** (0.05 and 0.01), indicating a stabilization phase.

Source:

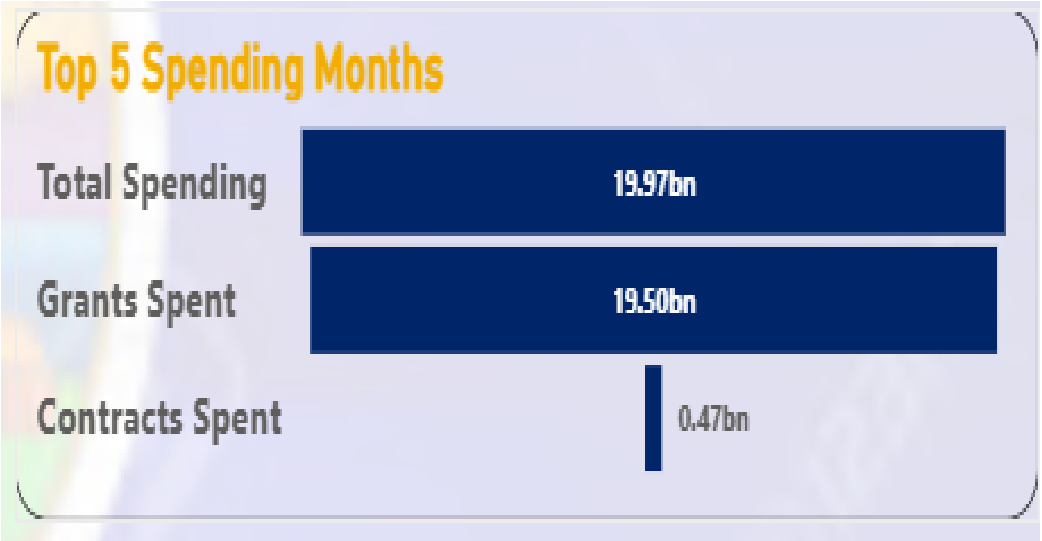
Business Value: This shows that federal spending is resilient and rebounds after economic or policy-driven downturns. Trend stability is crucial for long-term program planning and forecasting.

Actionable Recommendation: Agencies should prepare **contingency plans** for downturn years (like 2022) but capitalize on stabilization periods by applying for multi-year grants to lock in predictable funding.

Insight 2 - Total Funding Strongly Concentrated in Grants:

Insight Statement: The funnel chart shows that out of **$19.97B total spending**, **$19.50B** comes from Grants and only **$0.47B** from Contracts, reinforcing an extremely high dependency on grant-based funding.

Source:



Business Value: Such heavy reliance on grants means Kansas is vulnerable to federal policy changes, competitive grant cycles, and fluctuations in approval rates.

Actionable Recommendation: State agencies should actively explore contract-based opportunities, especially long-term federal procurement, to diversify funding and reduce risk from grant-cycle variability.

Insight 3 - Public Health, Coverage & Care Categories Drive Most Grant Allocations:

Insight Statement: The Sankey diagram shows that Public Health, Coverage & Care, and Human/Community categories have the strongest flows into Grants, indicating that these categories dominate Kansas's federal funding priorities.

Source:

Business Value: These top categories reveal where Kansas is most competitive and where federal government priorities align with state strengths — critical for future grant strategy.

Actionable Recommendation: Kansas should expand programs and partnerships in these dominant categories and train grant teams to target additional HHS initiatives in Public Health, Coverage & Care to maximize future award potential.

Supporting Citations:

Reference Textbook: Kolokolov, A., & Zelensky, M. (2021). Data Visualization with Microsoft Power BI: How to Design Savvy Dashboards. O'Reilly Media.

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